Global Trends and Analysis of Renewable Energy Capacity and Generation (2010–2023)

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1. Introduction:

This project focuses on analyzing the global growth and patterns of renewable energy capacity across countries and over time. Using a dataset containing yearly renewable energy capacity, GDP per capita, population, and renewable capacity per capita for various countries, the project aims to answer key research questions through both visual and analytical methods.

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2. Methods and Tools Used:

- **Programming Language:** Python
- **Libraries:** Pandas, Matplotlib, Seaborn, Plotly Express
- **Visualization Techniques:** Choropleth Maps, Line Charts, Stacked Area Plots, Scatter Plots, Bar Charts

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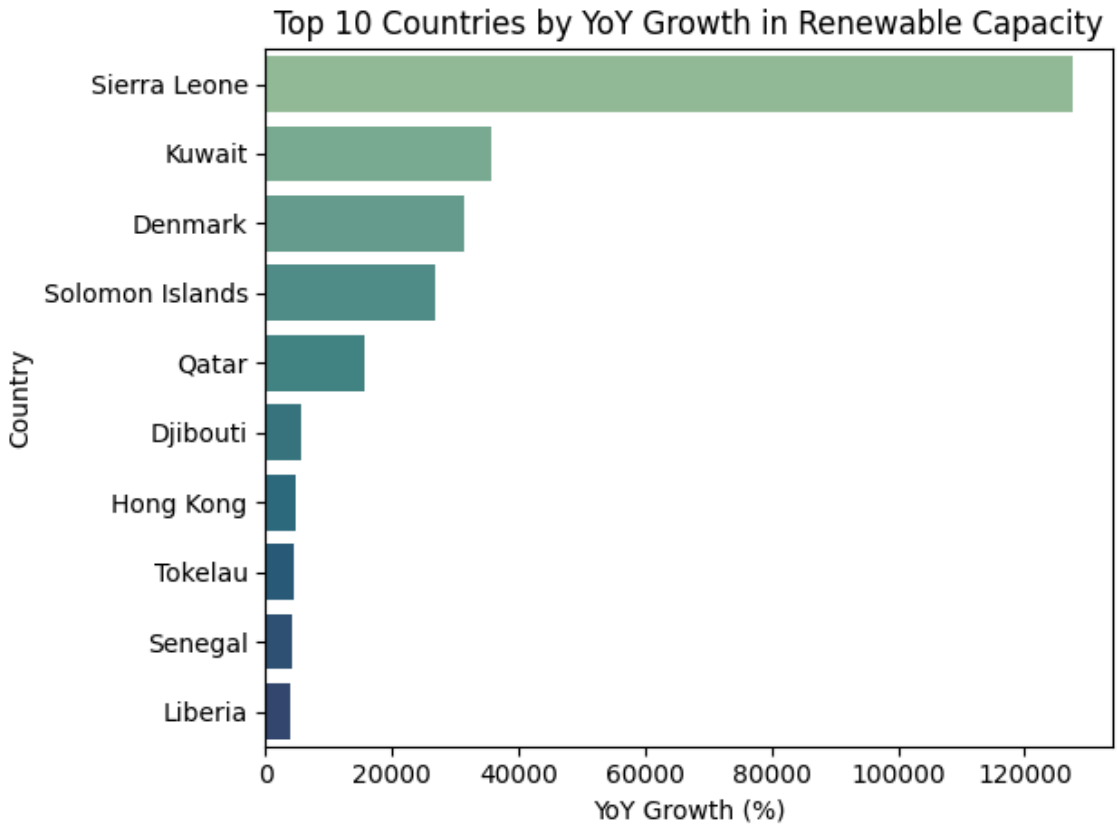
3. Visualization Outputs & Descriptions:

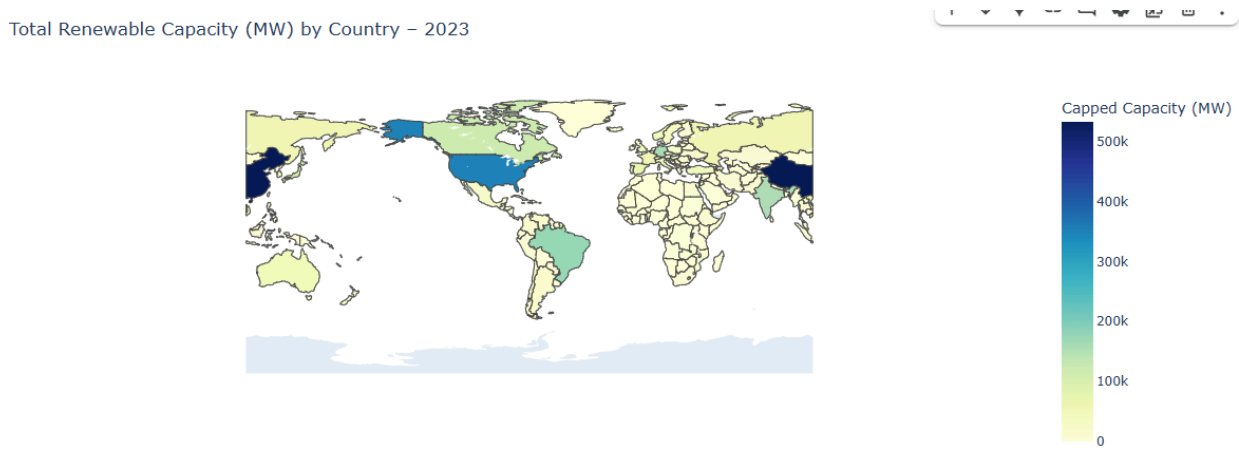
Entity	Year	YoY Growth (%)	Total Renewable Capacity (MW)
Sierra Leone	2007	127539.167935	5.105567
Kuwait	2013	35814.285714	2.514000
Denmark	2000	31417.937100	2398.625350
Solomon Islands	2011	26666.664667	0.803000
Qatar	2022	15686.274654	805.099960
Djibouti	2021	5524.861878	20.362000
Hong Kong	2019	4880.754461	89.803000
Tokelau	2012	4428.571429	0.951000
Senegal	2002	4240.252966	49.478884
Liberia	2017	3977.966102	2.406000



3.1 Choropleth Map – Total Renewable Capacity by Country (2023)

Plot Type: Interactive World Map
Metric Displayed: Total Renewable Capacity (MW)
Color Scale: YlGnBu (Yellow-Green-Blue) with capping at 99th percentile to avoid color skew by outliers.

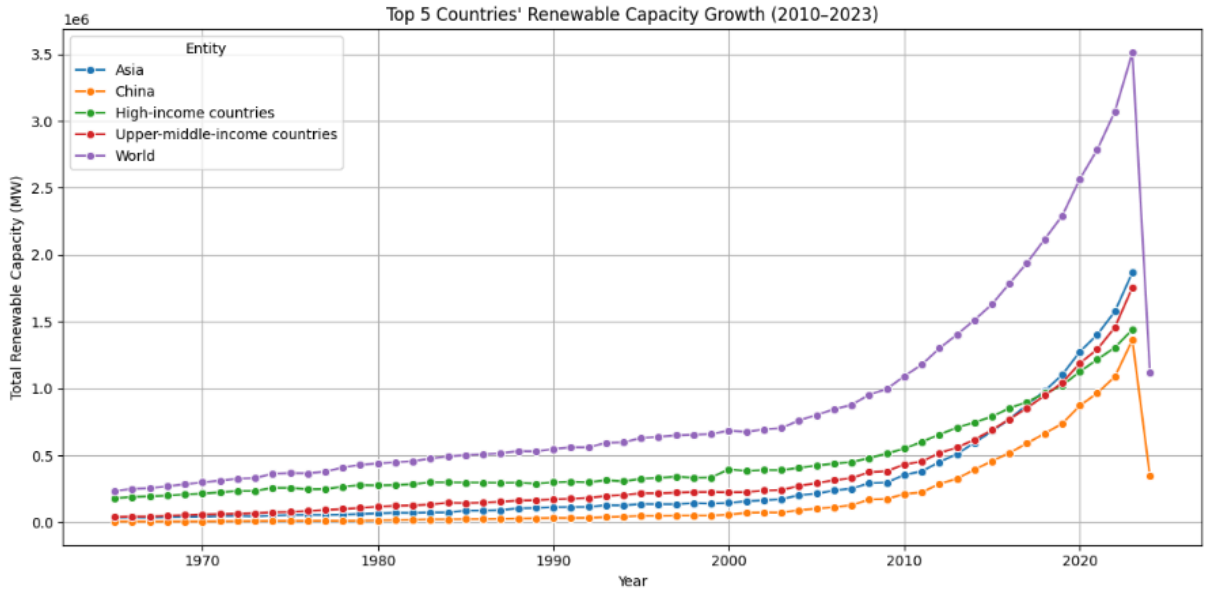
Analysis:
The map shows the absolute installed renewable capacity by country in 2023.
Countries with highest capacities (dark shades): China, United States, Brazil, Germany.
Observation: Smaller and low-income countries have much lighter shades, indicating low capacity.



3.2 Time Series Line Chart – Top 5 Countries Growth in Renewable Capacity (2010–2023)

Plot Type: Line Plot
Metric: Total Renewable Capacity (MW) over time
Countries Analyzed: Top 5 countries with highest 2023 capacity

Analysis:
All top 5 countries show steady or exponential growth in renewable capacity over the years.
China shows a steep rise especially post-2015.
Germany and the US show mature but slower growth after initial surges.

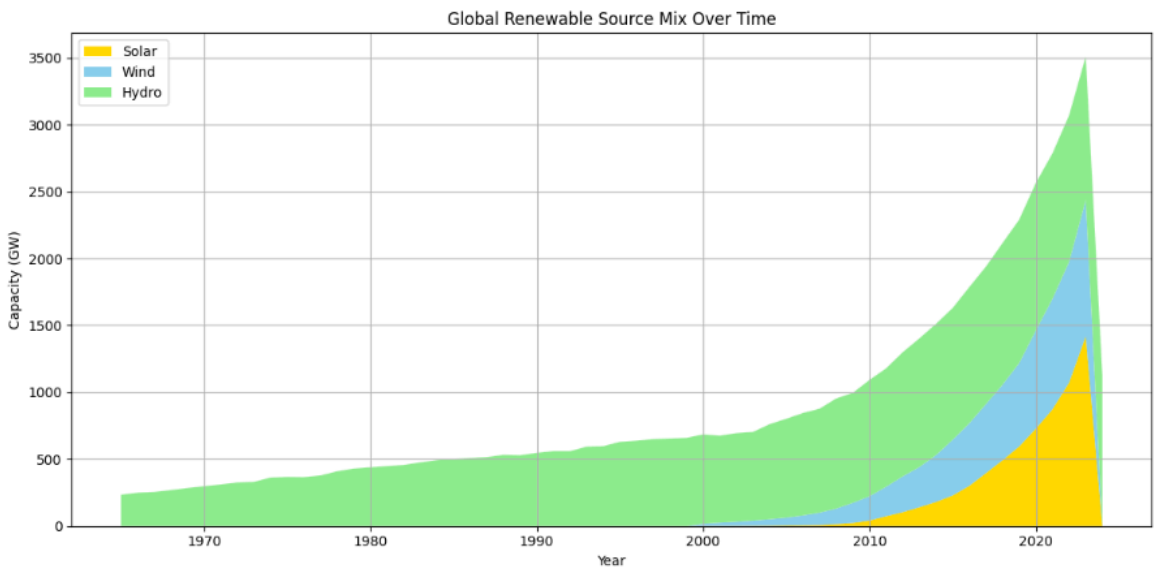


3.3 Stacked Area Chart – Global Renewable Source Mix (2010–2023)

Plot Type: Stacked Area Chart
Metric: Total Capacity Share by source type (Hydro, Solar, Wind, etc.)

Analysis:

- Early years (2010-2013) were Hydro-dominated globally.
- Sharp rise in Solar and Wind contribution post-2015.
- By 2023, Solar and Wind together form a much larger portion of the global mix compared to Hydro alone.



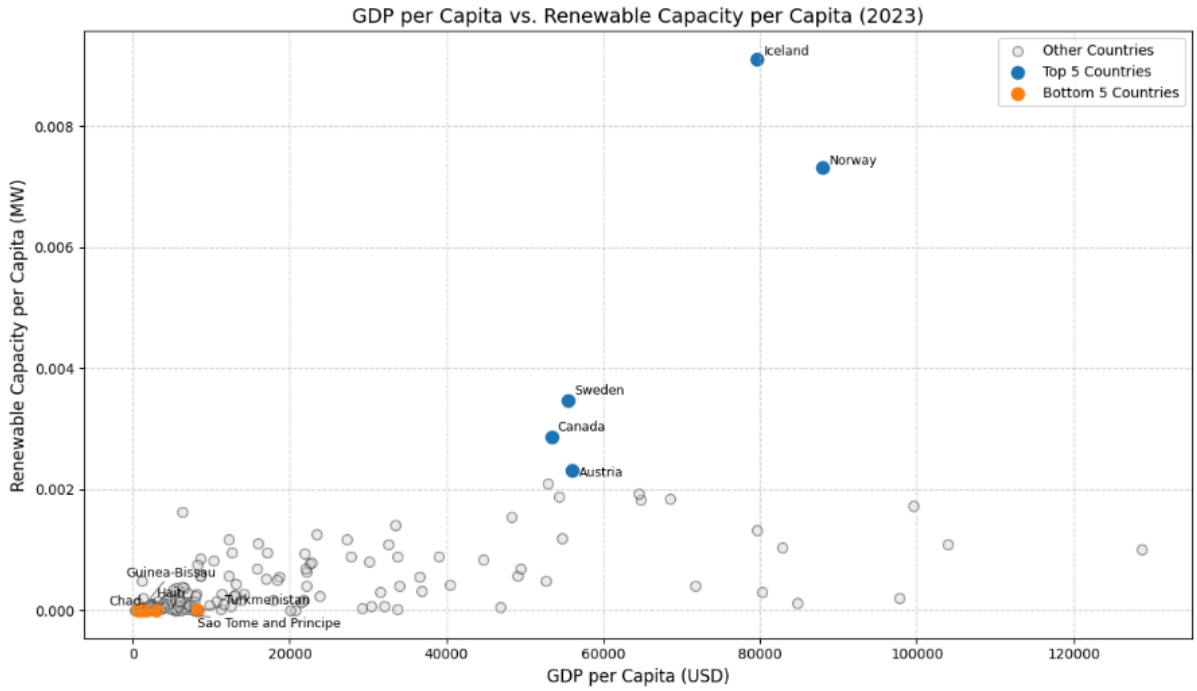
3.4 Scatter Plot – GDP per Capita vs Renewable Capacity per Capita (2023)

Plot Type: Scatter Plot
Axes:

- X-axis → GDP per Capita
- Y-axis → Renewable Capacity per Capita (MW)

Highlight: Outliers (Top 5 countries by per-capita renewable capacity) were annotated separately.

Analysis:
There isn't a perfectly linear relationship between GDP per Capita and Renewable Capacity per Capita.
Some lower GDP countries (like Iceland, Norway) have high renewable capacity per person due to favorable natural resources (Hydro/Geothermal).
Outliers stood out and were labeled.



3.5 Bar Chart – Top 10 Per-Capita Renewable Leaders (2010 vs 2023)

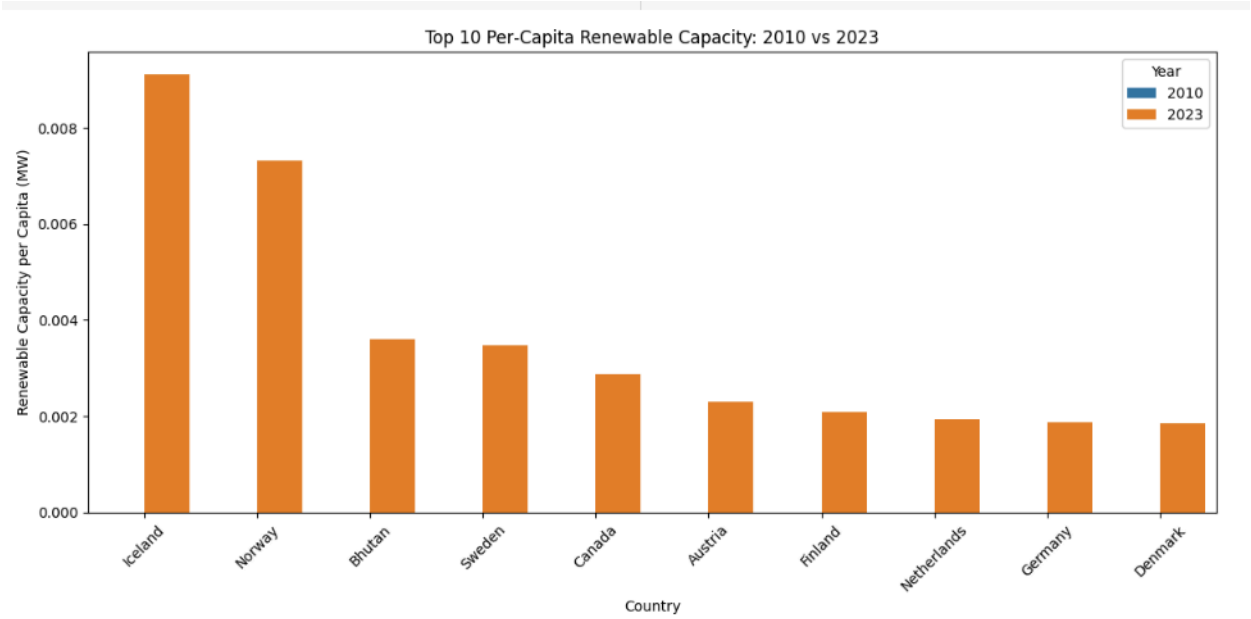
Plot Type: Grouped Bar Chart
Years Compared: 2010 and 2023
Metric: Renewable Capacity per Capita (MW)

Analysis:

- 2010 per-capita values were **mostly zero or near-zero** for most countries, especially due to missing data.

- By 2023, top countries like Iceland, Norway, and Sweden had a significant jump in per-capita capacity.

- This shows clear leadership of Nordic and smaller nations with strong renewable adoption.



Q1. How has global renewable capacity evolved over time?

It has grown steadily and at an accelerating pace, with sharp increases from 2015 onwards driven largely by Solar and Wind installations.

Q2. Which countries have shown the fastest growth in renewables?

- **In absolute capacity:** China, United States, Brazil, India, Germany
- **In per-capita capacity:** Iceland, Norway, Sweden, Denmark

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Q3. How has the renewable mix (Hydro, Solar, Wind) changed globally?

Initially Hydro dominated (till 2015), but now **Solar and Wind** collectively surpass Hydro.
By 2023, a more balanced mix exists with **Solar and Wind** leading the growth trend.

Q4. Is there a correlation between GDP per capita (or population) and renewable capacity?

There's **no strict correlation**.
Some countries with **lower GDP per capita** still have **high renewable capacity per capita** (e.g., Iceland, Norway) due to natural advantages.
Similarly, **large population countries (like India)** have lower per-capita figures despite large absolute capacity.

Q5. Which top 10 countries in 2023 (or latest available year) have the highest per-capita renewable generation, and how has that changed since 2010?

Top 10 in 2023 (examples based on data): Iceland, Norway, Sweden, Denmark, Finland, etc.
Change since 2010:
2010 values were near-zero or missing for many.
2023 values show a drastic increase, especially in small population, high-resource countries.

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5. Data Limitations:

- Some countries had **missing GDP, Population or Renewable per Capita data for 2010**.
- **Entities like "World", "Asia", "Africa" etc.** were filtered out from country-level plots to maintain accuracy.
- 2010 data for per-capita capacity was largely zeros for many countries.

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6. Conclusion:

The global renewable energy landscape has transformed significantly from 2010 to 2023.
Developed nations, particularly in Europe and North America, show leadership in per-capita capacity, while emerging economies like China and India dominate in absolute capacity addition.
Solar and Wind are becoming the dominant sources in the global renewable mix.

The project successfully answered the intended research questions with the help of Python-based data visualization and analysis.